

Week 3 – Design Studio

Design the support desk

The system. Customers submit support tickets through a web form. The ticket has two elements: the order, and a free-text message.

There are two models in this system:

- **A classifier**, a pre-trained binary logistic regression: given the ticket text, it returns a **score from 0 to 1** for "money is involved." It answers in milliseconds and is small enough to ship inside a function.
- **A language model**, the triage step: given the ticket and the account, it proposes
 - an **action** (answer · refund · hold · escalate)
 - a **refund amount**
 - a one-sentence **rationale** for the action
- **Decision**: the action. Refunds above the cap need a person. Nothing is refunded without the cap being checked.
- **Scale**: 50,000 tickets a month; a bad Monday is ten times an ordinary hour.
- **The customer**: a confirmation immediately; an answer within an hour.
- **The support team**: a queue of holds and escalations they can clear, with the rationale next to each.

Task. Develop the architecture for the system above: a rough idea of the scale the models must handle, then the system architecture itself.

1. **The three numbers.** Tickets per second on the bad Monday, the latency budget for the confirmation and for the answer, and the model cost per month with both models. Pick numbers and commit.
2. **The architecture.** Cover the full system. Name the tables. Draw the path of one ticket and mark each hop **sync** or **async**, with the reason. The two models need not live in the same place, and the classifier's score may decide what the language model sees. Include:
 - the web server
 - any databases
 - the two models
 - a queue mechanism for the manual review
 - anything else you need

3. **The failure path** (*if time*). What the customer and the support team see, and what runs, when the classifier is unavailable for an hour; and separately, when the language model is.